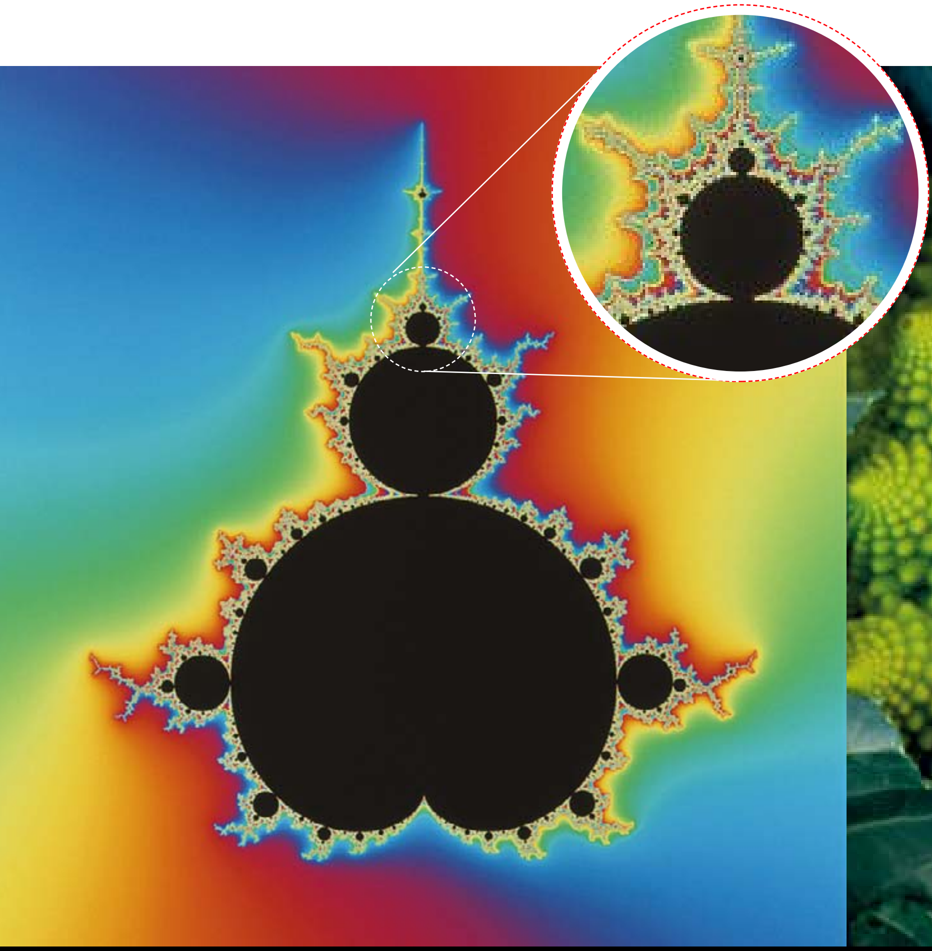
Freaky FRACTALS

Until about 100 years ago, mathematicians only studied *perfect* shapes like triangles and circles. But such shapes are rare in the real world. In nature, shapes are messy – think of a **wiggly coastline** or a **jagged mountain**. Unlike a circle, which gets smoother and flatter when you magnify it, a mountain stays just as jagged because you see ever more detail as you get closer. In 1975, mathematician Benoit Mandelbrot gave these *endlessly messy* shapes a name. He called them **fractals**.



Mandelbrot set
Mandelbrot created amazing fractal patterns on his computer by generating graphs with what mathematicians call "imaginary numbers". The most famous, called the Mandelbrot set, is said to be the most complex object in mathematics. It gets ever more detailed and beautiful when you zoom in and enlarge tiny portions of it, and the detail continues forever. Even stranger, the same basic shapes appear again and again, though with infinite variations.

Fractal broccoli
A fractal made of small copies of itself is said to be "self-similar". Broccolis and cauliflowers are self-similar fractals because the florets (and the tiny florets on them) are the same shape as the whole vegetable. Romanescu broccoli even looks a bit like the Mandelbrot set.